HAT-P-32b

R-0020

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_HATP-32_171220` · created 2026-07-09T07:55:27+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458107.71341 +/- 0.00093 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.156 +/- 0.0035
Transit depth (Rp/Rs)^2	2.43 +/- 0.11 [%]
Orbital Inclination (inc)	88.55 +/- 0.97
Airmass coefficient 1 (a1)	1.1608 +/- 0.0069
Airmass coefficient 2 (a2)	-0.1176 +/- 0.0026
Scatter in the residuals of the lightcurve fit is	0.6 %
Best Comparison Star	None
Optimal Aperture	4.68
Optimal Annulus	8.68
Transit Duration (day)	0.1306 +/- 0.0011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.156 ± 0.0035 vs 0.1489 (deviation 1.86σ)
Duration fit vs published	0.1306 d vs 0.1304 d (deviation 0.15σ)
Mid-time O–C	3.4 min
Depth SNR	40.9
Residual scatter	0.6 %

Light curve

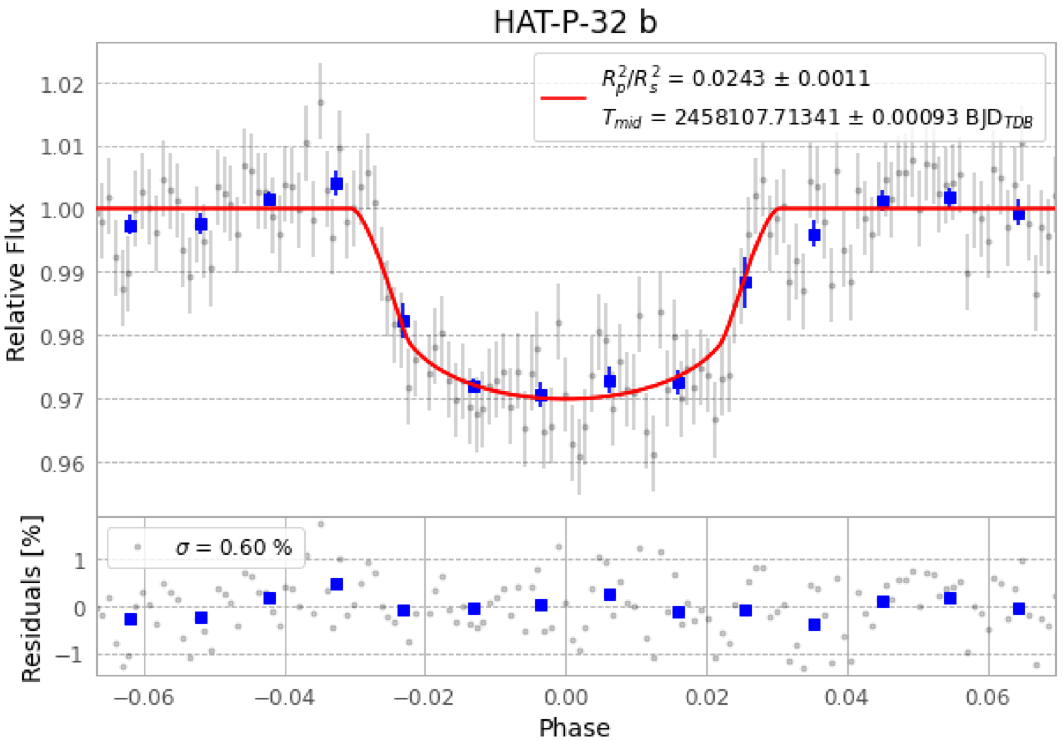

Figure 1 — FinalLightCurve_HAT-P-32 b_2017-12-19.png (SHA-256 832417d9aa58214a…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [423, 286], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 3062fc39f3a42b56…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13`

Data provenance

142 FITS frames (93 MB), `HATP-32171220013343.FITS` → `HATP-32171220083612.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-171220.log (34b5d5a88c67…), FinalLightCurve_HAT-P-32 b_2017-12-19.png (832417d9aa58…), FinalLightCurve_HAT-P-32 b_2017-12-19.csv (ad54aa0b901d…)

Compute resources

Wall time 150.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)